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1. Executive Summary

MolKit is a browser-native productivity platform that bundles 41 focused micro-tools into a single, cohesive experience. Every tool runs entirely client-side — no accounts, no data uploads to a server, and no paywalls. Users range from freelancers and developers to students and content creators who need practical, instant-access utilities without context-switching between dozens of separate web apps.

This document covers the full Product Requirements (PRD) and Software Requirements Specification (SRS) for building a new project based on MolKit's feature set, serving as the authoritative reference for design, engineering, and QA teams.

2. Product Overview

Product Name	MolKit (clone / new project)
Tagline	Pick a tool, do the job, keep moving.
Platform	Web (Browser-first, Progressive Web App optional)
Architecture	Static frontend (Next.js / React), zero backend for core tools
Total Tools	41 tools across 7 categories, grouped into 5 Kits
Monetisation	Free tier with optional Pro tier (premium tools / exports)
Target Users	Freelancers, developers, students, content creators, small business owners
Key Value Prop	All-in-one browser toolbox: private, instant, no login required

3. Goals & Success Metrics

Goal	KPI	Target
User Retention	Return visit rate	≥ 40% within 30 days
Tool Engagement	Avg tools used per session	≥ 2
Performance	LCP (Largest Contentful Paint)	< 2.5 s
Privacy	External API calls for core tools	0
Accessibility	WCAG 2.1 AA compliance	100% of core tools
Coverage	Tools launched at MVP	≥ 25 of 41

4. Feature Inventory – All 41 Tools

#	Tool Name	Category	Description
1	ATS Resume Builder	Planning	Build ATS-optimised resumes with 7 themes, keyword scoring, and PDF/HTML export
2	Text Pattern Extractor	Text	Extract emails, URLs, numbers, dates, and custom regex patterns from any text
3	Data Cleanup Suite	Text	Normalize, deduplicate, sort, and reformat messy text lists
4	CSV / Table Builder	Data	Edit tabular data visually; export as CSV, Markdown, JSON, HTML, or SQL
5	Regex Builder	Developer	Build, test, and debug regular expressions with live match highlighting
6	Structured Communication Builder	Planning	Generate follow-ups, inquiries, reminders, and cover letters from templates
7	Bulk File Naming	Text	Batch-rename filenames with prefixes, suffixes, counters, and date tokens
8	Markdown Editor	Text	Full-featured split-pane Markdown editor with live preview and export
9	Proposal Generator	Planning	Draft client proposals with scope, timeline, pricing, and next steps
10	API Request Tester	Developer	Send HTTP requests with headers, params, and JSON body from the browser
11	cURL Converter	Developer	Convert cURL commands to fetch/Axios snippets and vice-versa
12	Timezone Overlap	Planning	Find overlapping meeting windows across two or more time zones
13	Pricing Calculator	Numbers	Break down totals with flat amounts, percentages, and discount layers
14	Quotation Generator	Numbers	Build itemized quotes with totals, tax, validity date, and PDF export
15	Invoice Builder	Numbers	Create professional invoices with logo, line items, tax, and 6 templates
16	Currency Converter	Numbers	Convert between 160+ currencies with a live quick-reference rate table
17	Image Compressor	Media	Compress images client-side while preserving visual quality
18	Social Post Mockup Builder	Media	Preview social posts with profile UI, stats, comments per platform
19	QR Code Builder	Links	Generate branded QR codes with rich payloads and export options
20	Text Extractor (File to Text)	Text	Extract editable text from PDF, images, or plain-text files
21	Images to PDF	PDF	Combine multiple images into a single printable PDF
22	PDF to DOCX	PDF	Convert PDFs to editable Word documents with heading detection

23	Nginx Config Generator	Developer	Build complete Nginx server configs with SSL, proxy, and security headers
24	.htaccess Rules Generator	Developer	Create Apache redirect rules from simple toggles and form inputs
25	YouTube Video Inspector	Links	Parse YouTube links and enrich video metadata locally
26	JWT Decoder	Developer	Decode JWT header and payload locally in the browser
27	Diagram Builder	Developer	Build flowcharts, ERDs, and architecture maps with JSON editing
28	Food Costing Tool	Numbers	Calculate recipe costs, food cost %, and menu pricing
29	Sketch & Wireframe Tool	Media	Draw mockups and wireframes with a full SVG canvas editor
30	FocusFlow	Planning	Pomodoro timer and task manager in one focused workspace
31	Capstone Idea Builder	Planning	Generate a structured capstone concept from domain and problem input
32	Bulk Certificate Generator	Planning	Generate professional certificates in bulk from a name list
33	ID Card / Badge Generator	Media	Design and bulk-generate ID cards with photo, QR, and custom fields
34	Text Intelligence Analyzer	Text	Real-time readability, passive voice, keyword density, and quality scoring
35	Background Remover	Media	Remove image backgrounds via AI API or fully local processing
36	Icon System Builder	Media	Convert an image into a full favicon and icon package
37	System Spec Builder	Developer	Output agent-ready system prompts, Markdown specs, and JSON configs
38	SVG System Builder	Media	Build and export structured SVG icons and logos with live code output
39	UI System Blueprint	Planning	Browse 54 UI templates and export AI-ready design rules
40	Color System Toolkit	Media	Build color palettes, harmonies, gradients, and design tokens
41	Contract & Agreement Builder	Planning	Generate dynamic business and personal contracts with live preview

5. Feature Specifications (PRD per Tool)

Each specification below follows the format: Description → Key Sub-Features → User Stories → Technical Notes.

F-01 ATS Resume Builder

Planning

A guided, structured resume builder that outputs ATS-friendly resumes. Users fill in contact info, experience, education, skills, certifications, projects, languages, and volunteer sections. A live HTML preview updates in real-time across 7 visual themes.

Key Sub-Features

- 7 colour/layout themes selectable in real time
- ATS keyword match scoring against pasted job-description keywords
- Section drag-and-drop reordering
- Export to PDF, HTML, and plain text (.txt)
- Save/load draft via localStorage
- Custom sections for non-standard resume blocks

User Stories

- *As a job seeker I want to paste a job description and see which keywords I'm missing so I can improve my match score.*
- *As a user I want to reorder resume sections so that the most relevant content appears first.*
- *As a freelancer I want to export my resume as a PDF so I can send it immediately.*

Technical Notes

- All processing client-side — no resume data leaves the browser
- PDF export via browser print API or jsPDF
- LocalStorage for draft persistence (no account needed)

F-02 Text Pattern Extractor

Text

Extracts structured data patterns from any pasted text block using preset and custom regex rules. Supports emails, URLs, phone numbers, dates, IPv4 addresses, and user-defined patterns.

Key Sub-Features

- Preset pattern chips (Email, URL, Phone, Date, IP, Number)
- Custom regex input with live validation
- Extracted results displayed as a clean list with copy-all button
- Duplicate filtering toggle
- Export extracted results as CSV or plain text

User Stories

- *As a developer I want to extract all URLs from a blob of HTML so I can audit broken links.*
- *As an analyst I want to pull all dates from meeting notes so I can create a timeline.*

Technical Notes

- Uses the browser's native RegExp engine — zero library dependency
- Regex errors surfaced with human-readable messages

F-03 Data Cleanup Suite

Text

Normalises, deduplicates, sorts, trims, and reformats messy text lists. Designed for cleaning copied data from spreadsheets, emails, or databases.

Key Sub-Features

- Trim whitespace, remove blank lines, strip HTML tags
- Deduplicate (case-sensitive or insensitive)
- Sort alphabetically (A-Z / Z-A) or by line length
- Case conversion: UPPER, lower, Title Case, Sentence case
- Find & replace with regex support
- Number/bullet list wrapping

User Stories

- *As a data analyst I want to paste a messy CSV column and remove duplicates in one click.*
- *As an operations manager I want to normalise a list of names to Title Case before importing.*

Technical Notes

- All string operations run in-browser; supports large inputs (>10k lines)
- Operations can be chained and applied in sequence

F-04

CSV / Table Builder

Data

A visual spreadsheet-like editor for creating and editing tabular data. Users can add/remove rows and columns, rename headers, and export in multiple formats.

Key Sub-Features

- Inline cell editing with keyboard navigation (Tab/Arrow keys)
- Add/remove rows and columns dynamically
- Export as CSV, Markdown table, JSON array, HTML table, or SQL INSERT
- Import from pasted CSV text
- Column sorting and filtering

User Stories

- *As a developer I want to paste CSV data and export it as SQL INSERT statements for a seed file.*
- *As a writer I want to build a Markdown table without remembering the pipe syntax.*

Technical Notes

- Pure React state management — no external spreadsheet library required for MVP
- Large datasets (>1000 rows) may require virtual scrolling

F-05

Regex Builder

Developer

An interactive regex playground where users write patterns and test them against sample text with live highlighting. Includes a flag selector and match details panel.

Key Sub-Features

- Live match highlighting in the test string
- Match list with group captures shown per match
- Regex flag toggles (global, case-insensitive, multiline, dotAll)
- Common pattern library (email, URL, date, etc.)
- Export as JavaScript, Python, or PHP snippet

User Stories

- *As a developer I want to test my regex against sample data before adding it to production code.*

- *As a beginner I want to browse common patterns and understand what each part does.*

Technical Notes

- Uses browser RegExp — no server round-trip
- Error boundary catches invalid regex and shows descriptive messages

F-06 Structured Communication Builder

Planning

Generates professional written communications from structured templates. Covers follow-up emails, client inquiries, meeting reminders, and cover letters.

Key Sub-Features

- Template types: Follow-Up, Inquiry, Reminder, Cover Letter, Proposal Intro
- Dynamic placeholder fields (recipient name, company, role, deadline)
- Tone selector (Formal, Semi-formal, Friendly)
- Copy-to-clipboard and optional mailto: link generation
- Preview renders final composed text

User Stories

- *As a freelancer I want to generate a polite payment follow-up email in seconds.*
- *As a job applicant I want a cover letter scaffold tailored to a specific role.*

Technical Notes

- Template engine built with simple string interpolation — no external dependencies
- Future: AI-assisted tone rewriting via Anthropic API

F-07 Bulk File Naming

Text

Generates a list of renamed filenames given a base name and a set of rules: prefix, suffix, counter format, date token, and extension.

Key Sub-Features

- Base name + sequential number (configurable start, step, padding)
- Prefix and suffix text fields
- Date token insertion (YYYY-MM-DD, YYYYMMDD)
- Extension override
- Preview list of all generated names; copy as newline-separated list

User Stories

- *As a photographer I want to rename 200 files with a date prefix and sequential number.*
- *As a developer I want to generate asset filenames for a design handoff.*

Technical Notes

- Client-side string generation only — actual file renaming requires user to apply names manually or via OS tools
- No file access API (File System Access API could enable direct renaming in supported browsers)

F-08 Markdown Editor

Text

A split-pane Markdown editor with a live HTML preview pane. Supports the full CommonMark spec plus GFM extensions (tables, task lists, strikethrough).

Key Sub-Features

- Split view: editor left, rendered preview right

- Toolbar for common formatting (Bold, Italic, Heading, Link, Image, Table, Code block)
- Word count and character count in status bar
- Export as .md file or rendered HTML file
- Dark mode toggle

User Stories

- *As a technical writer I want to write documentation in Markdown and instantly see how it renders.*
- *As a student I want to export my notes as an HTML file to share with classmates.*

Technical Notes

- Rendering via marked.js or markdown-it library
- Syntax highlighting in the editor via CodeMirror or Monaco (lite)

F-09 Proposal Generator

Planning

A guided form that produces a clean, professional client proposal document with scope, deliverables, timeline, pricing, and next steps.

Key Sub-Features

- Project overview and problem statement fields
- Deliverables list with editable line items
- Timeline with milestone rows
- Pricing table with subtotal, tax, and total
- Client and agency signature block
- Export as PDF or HTML

User Stories

- *As a freelancer I want to generate a proposal for a new client in under 10 minutes.*
- *As a small agency I want a consistent proposal format across all team members.*

Technical Notes

- PDF generation via browser print dialog or jsPDF
- All data stays in localStorage — no server storage

F-10 API Request Tester

Developer

A lightweight in-browser HTTP client similar to Postman. Users configure method, URL, headers, query params, and body then send requests and inspect responses.

Key Sub-Features

- Method selector (GET, POST, PUT, PATCH, DELETE, HEAD, OPTIONS)
- Headers editor (key-value pairs)
- Query params editor (auto-appended to URL)
- JSON / form-data / raw body editor
- Response viewer: status, headers, pretty-printed JSON body
- Request history (localStorage)

User Stories

- *As a developer I want to test a REST endpoint without leaving my browser.*
- *As a QA engineer I want to quickly verify an API response shape.*

Technical Notes

- Uses the browser fetch() API — subject to CORS restrictions

- A CORS proxy toggle can route requests through a lightweight proxy for CORS-blocked endpoints
- Sensitive request data (auth tokens) never persisted to server

F-11 cURL Converter

Developer

Converts cURL commands to JavaScript fetch, Axios, Python requests, and PHP cURL snippets — and converts back from fetch to cURL.

Key Sub-Features

- Input: raw cURL command
- Output tabs: fetch, Axios, Python requests, PHP cURL
- Reverse mode: paste fetch → get cURL
- Copy button per output tab
- Syntax highlighting in output

User Stories

- *As a backend developer I want to copy a cURL from API docs and get a ready-to-paste fetch snippet.*
- *As a DevOps engineer I want to convert a fetch call into a cURL for terminal testing.*

Technical Notes

- Parser built with regex + AST approach for robust flag handling
- Handles --data-raw, --header, --user, --cookie, and common cURL flags

F-12 Timezone Overlap

Planning

Helps distributed teams find mutually acceptable meeting times by visualising the working hours of two or more time zones side by side.

Key Sub-Features

- Select up to 4 time zones from a searchable dropdown
- Visual 24-hour band showing overlap in green
- Working hours configuration per zone (default 9am–6pm)
- Current time indicator
- Copy overlap window as text (e.g., '2 PM – 4 PM UTC / 10 AM – 12 PM ET')

User Stories

- *As a remote team lead I want to see when all team members are online simultaneously.*
- *As a freelancer I want to schedule a call with a client in a different country.*

Technical Notes

- Uses Intl.DateTimeFormat and date-fns-tz for timezone calculations
- No server calls — all timezone data bundled client-side

F-13 Pricing Calculator

Numbers

A flexible calculator for building up a total price from base amounts, percentage add-ons (markup, VAT), and flat surcharges or discounts.

Key Sub-Features

- Base amount input
- Add unlimited line items: flat amount or % of running total
- Label each line item

- Running subtotal shown after each line
- Final total highlighted
- Copy result as text or export as PDF

User Stories

- *As a consultant I want to add a 15% markup then a 10% tax to my base project fee.*
- *As a shop owner I want to calculate my retail price from cost price with margin.*

Technical Notes

- Pure arithmetic — no external library
- Percentage applied to the subtotal at each step for compounding accuracy

F-14

Quotation Generator

Numbers

A form-driven tool that produces a formatted quotation document with client details, itemised services, totals, tax, and validity date.

Key Sub-Features

- Your company and client details
- Line items: description, quantity, unit price
- Tax rate field (applied to subtotal)
- Discount field (% or flat)
- Validity date
- Notes / terms field
- Export as PDF

User Stories

- *As a freelancer I want to send a professional quote to a potential client without using Word.*
- *As a small business I want to generate branded quotes with my logo.*

Technical Notes

- PDF via jsPDF or browser print
- Quote number auto-increments in localStorage

F-15

Invoice Builder

Numbers

Creates professional invoices with logo upload, itemised line items, multiple tax rates, payment details, and 6 visual templates.

Key Sub-Features

- Logo upload (rendered in invoice header)
- 6 invoice templates (colour schemes and layouts)
- Line items: description, qty, rate, amount
- Multiple tax rate rows
- Payment details section (bank, PayPal, etc.)
- Invoice number, date, due date fields
- Export PDF / HTML

User Stories

- *As a freelance designer I want to generate and download a branded invoice in under 2 minutes.*
- *As a contractor I want to track invoice numbers locally without a full accounting app.*

Technical Notes

- Template rendering via React component + CSS print stylesheet
- Logo stored as base64 data URL in localStorage

F-16 Currency Converter

Numbers

Converts amounts between 160+ world currencies using up-to-date exchange rates, plus a quick-reference multi-currency table.

Key Sub-Features

- Amount input with source and target currency selectors
- Quick-reference rate table: 1 unit of base vs. 10+ common currencies
- Rate source attribution and last-updated timestamp
- Swap currencies button
- Historical rate note

User Stories

- *As a traveller I want to quickly know how much €200 is in Thai Baht.*
- *As a freelancer I want to convert my USD invoice total to GBP for a UK client.*

Technical Notes

- Exchange rates fetched from a free public API (e.g., exchangerate-api.com) with 1-hour client-side cache
- Fallback to static rates if API is unavailable

F-17 Image Compressor

Media

Compresses images (JPEG, PNG, WebP) in the browser using canvas re-encoding, reducing file size while controlling quality via a slider.

Key Sub-Features

- Drag-and-drop or click-to-upload
- Quality slider (1–100)
- Output format selector (JPEG, PNG, WebP)
- Before/after file size and dimension comparison
- Batch upload (up to 10 images)
- Download compressed image(s)

User Stories

- *As a blogger I want to compress my hero image before uploading to my CMS.*
- *As a developer I want to batch-compress screenshots for a pull request.*

Technical Notes

- Canvas API re-encoding — no server upload
- WebP output requires browser support check with JPEG fallback

F-18 Social Post Mockup Builder

Media

Previews social media posts as they would appear on Twitter/X, LinkedIn, Instagram, and Facebook, with configurable profile details, like counts, and comments.

Key Sub-Features

- Platform selector: Twitter/X, LinkedIn, Instagram, Facebook
- Editable profile name, handle, avatar

- Post text input with character counter per platform limit
- Image attachment toggle
- Fake engagement stats (likes, comments, shares)
- Export mockup as PNG

User Stories

- *As a social media manager I want to preview a post before publishing to catch formatting issues.*
- *As a trainer I want to create realistic-looking social screenshots for course materials.*

Technical Notes

- DOM-to-image export via html2canvas
- Character limits enforced per platform

F-19 QR Code Builder

Links

Generates QR codes for URLs, text, vCard, WiFi credentials, SMS, email, and phone numbers, with colour and logo customisation.

Key Sub-Features

- Payload types: URL, text, vCard, WiFi, SMS, Email, Phone
- Foreground and background colour pickers
- Logo image overlay (centred)
- Error correction level selector (L / M / Q / H)
- Size and margin controls
- Export as PNG or SVG

User Stories

- *As a restaurant owner I want a QR code linking to my menu with my logo in the centre.*
- *As an event organiser I want to generate WiFi QR codes for guest badges.*

Technical Notes

- QR generation via qrcode.js or similar zero-dependency library
- SVG export for print-quality output

F-20 Text Extractor

Text

Extracts readable text from uploaded files including plain text, PDFs (text layer), and images (via browser OCR or Tesseract.js).

Key Sub-Features

- Supported inputs: .txt, .md, .csv, .pdf (text layer), .png, .jpg, .webp
- PDF text extraction via pdf.js
- Image OCR via Tesseract.js (runs locally in Web Worker)
- Extracted text displayed in editable textarea
- Copy and download as .txt

User Stories

- *As a researcher I want to copy text from a scanned image without retyping it.*
- *As an analyst I want to extract text from a PDF report into a plain text file.*

Technical Notes

- Tesseract.js WASM runs in a Web Worker to avoid blocking the main thread
- Large PDFs processed page by page with progress indicator

F-21**Images to PDF**

PDF

Combines multiple uploaded images into a single PDF document with configurable page size, orientation, and image fit.

Key Sub-Features

- Multi-file drag-and-drop upload
- Reorder images before combining (drag handles)
- Page size: A4, Letter, image-native
- Image fit: Fit, Fill, Stretch
- Margin controls
- Download combined PDF

User Stories

- *As a student I want to scan multiple pages and combine them into one PDF to submit.*
- *As a photographer I want to create a PDF portfolio from a set of JPEGs.*

Technical Notes

- PDF generation via jsPDF or pdf-lib — fully client-side
- Large image sets may require chunked processing

F-22**PDF to DOCX**

PDF

Converts uploaded PDF files to editable Word documents (.docx), with heading detection and basic formatting preservation.

Key Sub-Features

- PDF upload (drag-and-drop or click)
- Heading and paragraph detection
- Table extraction (best-effort)
- Download .docx file

User Stories

- *As a student I want to edit a PDF lecture slide in Word.*
- *As an admin I want to convert a scanned form to an editable document.*

Technical Notes

- Uses pdf.js for text extraction and docx npm library for DOCX generation
- Scanned PDFs require OCR pre-processing (link to Text Extractor tool)

F-23**Nginx Config Generator**

Developer

A form-driven generator for complete Nginx server block configurations, covering static sites, reverse proxies, SSL, gzip, and security headers.

Key Sub-Features

- Server type: static site, reverse proxy, load balancer
- SSL / HTTPS toggle (with Let's Encrypt cert path fields)
- Gzip compression toggle
- Security headers block (CSP, HSTS, X-Frame-Options)
- Rate limiting block

- WWW-to-non-www redirect toggle
- One-click copy output

User Stories

- *As a sysadmin I want to generate a production-ready Nginx config without looking up syntax.*
- *As a developer I want to add security headers to my site without manual research.*

Technical Notes

- Pure string template generation — no server needed
- Output validated against known Nginx directive syntax rules

F-24 .htaccess Rules Generator

Developer

Creates Apache .htaccess rewrite rules from simple toggle inputs: HTTPS redirect, www redirect, trailing slash, custom 404, and more.

Key Sub-Features

- HTTPS redirect rule
- WWW / non-WWW redirect
- Trailing slash enforcement
- Custom 404 error page
- Directory listing disable
- Hotlink protection
- Password protect directory (basic auth scaffold)

User Stories

- *As a web developer I want to generate redirect rules without memorising mod_rewrite syntax.*
- *As a site owner I want to force HTTPS with a single toggle.*

Technical Notes

- String template output; includes comments explaining each rule
- Validation alerts for conflicting rules

F-25 YouTube Video Inspector

Links

Parses a YouTube URL and displays structured video metadata: video ID, thumbnail URLs, embed code, and available resolutions.

Key Sub-Features

- Extract video ID from any YouTube URL format
- Display thumbnail URLs (default, medium, high, maxres)
- Generate embed iframe code with configurable options
- Detect playlist vs. single video
- Copy thumbnail URL or embed code

User Stories

- *As a blogger I want the maxres thumbnail URL for a YouTube video to use as a post image.*
- *As a developer I want the embed code with autoplay disabled.*

Technical Notes

- YouTube URL parsing is fully local (regex-based)
- Metadata enrichment via YouTube oEmbed API (public, no API key needed for basic info)

F-26**JWT Decoder**

Developer

Decodes a JSON Web Token (JWT) locally in the browser, displaying the header, payload, and signature, with expiry check.

Key Sub-Features

- Paste JWT input
- Decoded header and payload displayed as pretty-printed JSON
- Signature section shown (not verified unless secret provided)
- Expiry (exp claim) highlighted with relative time (expired / valid for X hours)
- Copy decoded JSON blocks

User Stories

- *As a developer I want to inspect a JWT token to debug authentication issues.*
- *As a security reviewer I want to check token claims without sending them to a third-party site.*

Technical Notes

- Decoding via `atob()` and `JSON.parse` — fully browser-native
- Signature verification optional (HMAC secret input)

F-27**Diagram Builder**

Developer

A visual diagram editor supporting flowcharts, ERDs, sequence diagrams, and architecture maps. Supports both drag-and-drop visual editing and JSON-based editing.

Key Sub-Features

- Shape library: process, decision, start/end, database, server, user
- Connectors with labels and arrow styles
- JSON view for power users
- Export as SVG or PNG
- Zoom, pan, grid snap

User Stories

- *As an architect I want to draw a system architecture diagram and export it for a presentation.*
- *As a developer I want to sketch an ERD to plan my database schema.*

Technical Notes

- Built on React Flow or a lightweight custom SVG canvas
- JSON schema enables reproducible diagrams via URL hash or file import

F-28**Food Costing Tool**

Numbers

Calculates recipe costs, food cost percentage, and menu pricing for restaurants and food businesses.

Key Sub-Features

- Ingredient list with unit, quantity, and cost per unit
- Automatic total recipe cost calculation
- Food cost % input (target) → suggested menu price
- Portion/yield adjustment
- Save and load recipes via `localStorage`
- Export as PDF report

User Stories

- *As a restaurant owner I want to know the exact cost per portion of my signature dish.*
- *As a caterer I want to calculate how much to charge per head for an event.*

Technical Notes

- Arithmetic only — no external dependencies
- Recipes stored in localStorage as JSON

F-29 Sketch & Wireframe Tool

Media

A full-featured SVG canvas editor for drawing interface mockups, wireframes, and free-form diagrams.

Key Sub-Features

- Drawing tools: rectangle, ellipse, line, arrow, text, image placeholder
- Select, move, resize, and rotate elements
- Snap to grid
- Layers panel
- Export as SVG or PNG
- Undo / redo stack

User Stories

- *As a UX designer I want to sketch a wireframe to share with a developer.*
- *As a product manager I want to mock up a screen flow for a stakeholder review.*

Technical Notes

- Built on Fabric.js or a custom SVG canvas implementation
- SVG export is lossless and scalable

F-30 FocusFlow

Planning

A combined Pomodoro timer and task manager. Tasks are organised into a queue; the timer runs per task; completed sessions are logged.

Key Sub-Features

- Pomodoro timer (configurable work/break durations)
- Task queue: add, reorder, mark complete
- Session log (date, task, duration)
- Desktop notification on timer end
- Ambient sound options (white noise, rain, etc.)
- Dark mode

User Stories

- *As a developer I want to work in focused 25-minute sprints on a task list.*
- *As a student I want to track how many Pomodoros I spend on each subject.*

Technical Notes

- Web Audio API for ambient sounds
- Notification API for timer alerts
- State persisted in localStorage

F-31 Capstone Idea Builder

Planning

Generates a structured academic capstone or thesis project concept from user-provided domain, problem context, and target audience.

Key Sub-Features

- Domain selector (IT, Business, Education, Health, etc.)
- Problem statement textarea
- Target audience field
- Output: project title, objectives, scope, methodology suggestion, expected outcomes
- Copy / download as text or PDF

User Stories

- *As a student I want a structured capstone idea generated from my vague problem description.*
- *As an instructor I want to generate example capstone topics to share with students.*

Technical Notes

- Template-based generation for MVP; AI-powered generation (Claude API) as a Pro feature

F-32 Bulk Certificate Generator

Planning

Generates printable certificates in bulk from a template and a name list, with custom fields like course name, date, and signature.

Key Sub-Features

- Name list input (paste or CSV upload)
- Certificate template selector (3+ templates)
- Custom text fields: title, course, date, signatory
- Logo and signature image upload
- Batch export as ZIP of PNGs or single multi-page PDF

User Stories

- *As a trainer I want to generate 50 certificates of completion after a workshop.*
- *As an event organiser I want to print attendance certificates for all participants.*

Technical Notes

- Canvas API for rendering each certificate
- JSZip for batch ZIP export
- jsPDF for multi-page PDF export

F-33 ID Card / Badge Generator

Media

Designs and bulk-generates ID cards and event badges with photo, QR code, and custom fields.

Key Sub-Features

- Card template designer (front/back)
- Custom fields: name, role, department, ID number, QR payload
- Photo upload per card or bulk via CSV+photo folder
- QR code per card (links to profile or check-in URL)
- Batch export as PNG or PDF

User Stories

- *As an HR manager I want to print employee ID cards with photos and department.*
- *As an event organiser I want to generate 200 attendee badges with QR check-in codes.*

Technical Notes

- Canvas API for card rendering
- CSV + photo association requires File System Access API or ZIP upload

F-34 Text Intelligence Analyzer

Text

Analyses pasted text for readability (Flesch-Kincaid), passive voice ratio, sentence length distribution, keyword density, and an overall quality score.

Key Sub-Features

- Flesch-Kincaid readability grade and reading ease score
- Passive voice sentence detection and percentage
- Average and max sentence length
- Top keyword frequency table
- Adverb and filler word count
- Overall writing quality score (0–100)

User Stories

- *As a content writer I want to know if my article is written at the right reading level.*
- *As a student I want to reduce passive voice in my essay before submitting.*

Technical Notes

- All NLP computations done with vanilla JS string parsing
- No external NLP library required for MVP; spaCy-style features as future enhancement

F-35 Background Remover

Media

Removes the background from uploaded images using an AI model or a local browser-based approach.

Key Sub-Features

- Upload image (PNG, JPG, WebP)
- Processing mode: AI API (higher quality) or Local (browser ML model)
- Transparent PNG output
- Preview on white and dark backgrounds
- Download result

User Stories

- *As a seller I want to remove the background from a product photo for my online store.*
- *As a designer I want to quickly cut out a subject from a photo.*

Technical Notes

- Local mode via @imgly/background-removal WASM model
- API mode via remove.bg or similar (requires API key from user)

F-36 Icon System Builder

Media

Converts a single uploaded image into a complete favicon and app icon package ready for web and mobile deployment.

Key Sub-Features

- Upload source image (SVG or high-res PNG)
- Generate sizes: 16, 32, 48, 64, 128, 180, 192, 512 px

- Output formats: PNG, ICO, SVG
- HTML link snippet for each size
- Download as ZIP

User Stories

- *As a developer I want to generate all favicon sizes from my logo in one step.*
- *As a designer I want to export an app icon set for a client handoff.*

Technical Notes

- Canvas API resizing + JSZip for bundle download
- ICO format requires a JS ICO encoder library

F-37 System Spec Builder

Developer

A visual configuration engine that outputs agent-ready system prompts, Markdown specification documents, and JSON configuration objects.

Key Sub-Features

- Guided form: agent name, role, capabilities, constraints, tone, output format
- Output tabs: System Prompt (text), Markdown Spec, JSON Config
- Copy and download each output format
- Template presets (customer support bot, coding assistant, etc.)

User Stories

- *As a developer I want to generate a system prompt for an AI agent without writing it from scratch.*
- *As a product manager I want to produce a Markdown spec for a new feature.*

Technical Notes

- Template-based string composition
- Claude API integration for AI-enhanced prompt refinement (Pro feature)

F-38 SVG System Builder

Media

A workspace for creating, editing, and exporting SVG icons and logos with a visual editor and live SVG code output.

Key Sub-Features

- Shape primitives: path, rect, circle, polygon, text
- Colour fill and stroke pickers
- Live SVG code panel (editable)
- Artboard size presets (16, 24, 32, 48, 512 px)
- Export as SVG or PNG at any DPI

User Stories

- *As an icon designer I want to build a clean SVG icon and get the optimised code.*
- *As a developer I want to tweak an SVG icon's colour and size without opening Figma.*

Technical Notes

- SVG DOM manipulation via vanilla JS or Snap.svg
- SVGO optimisation pass on export

F-39 UI System Blueprint

Planning

A reference engine with 54 UI templates that users can browse and export as AI-ready design rules for prompting design tools or LLMs.

Key Sub-Features

- 54 UI component/page templates (cards, navbars, modals, forms, dashboards, etc.)
- Searchable and filterable template gallery
- View template specifications: spacing, colours, typography rules
- Export as Markdown design brief or JSON design tokens
- Copy AI prompt for each template

User Stories

- *As a developer I want a design system reference to build consistent UI components.*
- *As a designer I want to export design tokens from a template to use in Figma.*

Technical Notes

- Static template data stored as JSON
- Export functions produce structured Markdown or JSON — no server

F-40

Color System Toolkit

Media

Builds semantic colour palettes, harmonic colour sets, gradients, and design tokens with live accessibility (WCAG) contrast checks.

Key Sub-Features

- Colour picker with HEX / RGB / HSL input
- Palette generator: complementary, triadic, analogous, split-complementary
- Tint and shade scale (50–950 Tailwind-style)
- Gradient builder (linear, radial) with CSS output
- WCAG contrast checker (AA/AAA badge per pair)
- Export as CSS custom properties, JSON design tokens, or Tailwind config

User Stories

- *As a designer I want to build a full colour system with accessible contrast ratios.*
- *As a developer I want to export my brand colours as Tailwind CSS config.*

Technical Notes

- Colour math via chroma.js or custom HSL arithmetic
- WCAG contrast ratio calculated per WCAG 2.1 formula

F-41

Contract & Agreement Builder

Planning

Generates dynamic, customisable business and personal contracts with a live preview. Contract types include freelance, NDA, service agreement, and lease.

Key Sub-Features

- Contract type selector: Freelance, NDA, Service Agreement, Employment, Lease
- Dynamic fields: parties, dates, terms, payment, jurisdiction
- Clause library for common additions
- Live document preview
- Export as PDF or .docx

User Stories

- *As a freelancer I want to generate a service agreement without paying a lawyer for a simple project.*

- *As a landlord I want a basic lease agreement template I can fill in and download.*

Technical Notes

- Template string interpolation for contract generation
 - Legal disclaimer: documents are templates, not legal advice
 - DOCX export via docx.js library
-

6. Tool Kits

Tools are grouped into curated Kits for common workflow contexts. A Kit is a collection page linking to 4–7 related tools.

Freelance & Business (6 tools)

- Invoice Builder
- Quotation Generator
- Proposal Generator
- Structured Communication Builder
- Pricing Calculator
- Contract & Agreement Builder

Certificates & Documents (4 tools)

- Bulk Certificate Generator
- ID Card / Badge Generator
- ATS Resume Builder
- PDF to DOCX

Developer Tools (7 tools)

- API Request Tester
- cURL Converter
- Regex Builder
- JWT Decoder
- Nginx Config Generator
- .htaccess Rules Generator
- System Spec Builder

Data & Text (5 tools)

- CSV / Table Builder
- Data Cleanup Suite
- Text Pattern Extractor
- Bulk File Naming
- Text Intelligence Analyzer

Media & Assets (7 tools)

- Image Compressor
- Background Remover
- QR Code Builder
- Icon System Builder
- Social Post Mockup Builder
- SVG System Builder
- Color System Toolkit

7. Software Requirements Specification (SRS)

7.1 System Overview

MolKit is a client-side web application built with React (Next.js) and deployed on a static CDN. All core tool logic executes in the browser; no user data is transmitted to a server. Optional integrations (currency rates, YouTube oEmbed) call public APIs without authentication.

7.2 Functional Requirements

FR-ID	Requirement	Priority	Tools
FR-01	All core tools must operate without an internet connection (offline-first)	Must	All 41
FR-02	No user data shall be transmitted to any server for core tool operations	Must	All 41
FR-03	Each tool must be accessible via a unique URL path (/tools/{slug})	Must	All 41
FR-04	Tool state must persist across page refresh via localStorage	Should	All stateful tools
FR-05	All export functions must produce files downloadable without a server	Must	22 tools
FR-06	Currency rates may be fetched from a public API with a 1-hour cache	Should	Currency Converter
FR-07	OCR processing must run in a Web Worker to prevent UI blocking	Must	Text Extractor
FR-08	Bulk operations must show a progress indicator for >10 items	Should	Bulk Certificate, ID Card
FR-09	Each tool must display a description and usage tips on first visit	Should	All 41
FR-10	Kit pages must aggregate related tools with descriptions and links	Must	5 Kits
FR-11	The tools index page must support search and category filtering	Must	Tools Index
FR-12	All tools must be keyboard-navigable and screen-reader compatible	Must	All 41

7.3 Data Requirements

No user accounts are required. All persistent data is stored in the browser via localStorage or IndexedDB. Sensitive data (resume content, invoices, contracts) never leaves the device. Optional Pro features may require a lightweight account for cloud sync.

7.4 Interface Requirements

- Responsive design: tools functional on screens ≥ 360 px wide
- Navigation: persistent top bar with logo, tool search, and Kit links
- Tool card grid on index page with category chips and search input
- Each tool page: breadcrumb navigation, tool title, description, and action area
- Dark mode support across all tool pages
- Loading skeleton states for tools with deferred asset loading (Tesseract, ML models)

8. Non-Functional Requirements

Category	Requirement	Metric
Performance	Initial page load (tools index)	LCP < 2.5 s on 4G
Performance	Tool interaction response	< 100 ms for UI updates
Performance	PDF export generation	< 3 s for standard documents
Performance	OCR processing (A4 image)	< 15 s in Web Worker
Scalability	Concurrent users (static CDN)	Unlimited (CDN-scaled)
Reliability	Uptime target	99.9% (Vercel/CDN SLA)
Security	Data transmission for core tools	Zero (fully client-side)
Security	Content Security Policy	Strict CSP headers on all pages
Security	API keys for Pro features	User-supplied; stored in sessionStorage only
Accessibility	WCAG standard	2.1 Level AA for all tools
SEO	Tool pages indexed	All 41 tool pages with meta tags and canonical URLs
Browser Support	Minimum supported browsers	Chrome 90+, Firefox 88+, Safari 14+, Edge 90+
Internationalisation	UI language for MVP	English; i18n architecture ready for v2
Privacy	Analytics	Privacy-first (Plausible or Fathom, no cookies)

9. Implementation Plan

9.1 Tech Stack

Layer	Technology	Rationale
Frontend Framework	Next.js 14 (App Router)	Static export, SSG, built-in routing
UI Library	React 18 + TypeScript	Component reuse, type safety
Styling	Tailwind CSS + shadcn/ui	Rapid UI development, accessible primitives
State Management	Zustand + localStorage	Lightweight, persistent state per tool
PDF Generation	jsPDF + @react-pdf/renderer	Client-side PDF creation
OCR	Tesseract.js (WASM)	Browser-native OCR, no server required
QR Codes	qrcode.js	Zero-dependency, SVG output
Canvas Ops	Fabric.js	Sketch tool, ID card canvas rendering
Diagrams	React Flow	Flowchart and ERD builder
Charts	Recharts	Data visualisation within tools
ML / BG Remove	@imgly/background-removal	WASM-based background removal
Icons	Lucide React	Consistent icon system
Colour Math	chroma.js	Palette generation and contrast
Deployment	Vercel (Static Export)	CDN, edge functions for future Pro API
Analytics	Plausible Analytics	Privacy-first, no cookies

9.2 Phased Delivery Roadmap

Phase 1 – Foundation (Weeks 1–3)

- Project scaffold: Next.js + Tailwind + shadcn/ui setup
- Design system: colours, typography, spacing tokens
- Navigation, tool index page with search + filter
- Kit pages structure
- Deploy to Vercel with CI/CD pipeline

Phase 2 – Text & Data Tools (Weeks 4–6)

- F-02 Text Pattern Extractor
- F-03 Data Cleanup Suite
- F-04 CSV / Table Builder
- F-05 Regex Builder
- F-07 Bulk File Naming
- F-08 Markdown Editor
- F-34 Text Intelligence Analyzer

Phase 3 – Planning & Business Tools (Weeks 7–9)

- F-01 ATS Resume Builder
- F-06 Structured Communication Builder
- F-09 Proposal Generator
- F-12 Timezone Overlap
- F-30 FocusFlow (Pomodoro)
- F-39 UI System Blueprint
- F-41 Contract & Agreement Builder

Phase 4 – Numbers & Finance Tools (Weeks 10–11)

- F-13 Pricing Calculator
- F-14 Quotation Generator
- F-15 Invoice Builder
- F-16 Currency Converter (with API integration)
- F-28 Food Costing Tool

Phase 5 – Developer Tools (Weeks 12–14)

- F-10 API Request Tester
- F-11 cURL Converter
- F-23 Nginx Config Generator
- F-24 .htaccess Rules Generator
- F-25 YouTube Video Inspector
- F-26 JWT Decoder
- F-37 System Spec Builder

Phase 6 – Media & Visual Tools (Weeks 15–18)

- F-17 Image Compressor
- F-18 Social Post Mockup Builder
- F-19 QR Code Builder
- F-29 Sketch & Wireframe Tool (Fabric.js)
- F-27 Diagram Builder (React Flow)
- F-35 Background Remover (WASM ML)
- F-36 Icon System Builder
- F-38 SVG System Builder
- F-40 Color System Toolkit

Phase 7 – PDF & Document Tools (Weeks 19–21)

- F-20 Text Extractor (Tesseract.js OCR)
- F-21 Images to PDF
- F-22 PDF to DOCX
- F-31 Capstone Idea Builder
- F-32 Bulk Certificate Generator
- F-33 ID Card / Badge Generator

Phase 8 – Polish & Launch (Weeks 22–24)

- Accessibility audit (WCAG 2.1 AA) across all 41 tools
- Performance optimisation: lazy loading, code splitting per tool

- SEO: meta tags, structured data, sitemap
- Cross-browser testing (Chrome, Firefox, Safari, Edge)
- Error monitoring (Sentry)
- Privacy analytics (Plausible)
- Public launch + Product Hunt submission

10. Risk Register

Risk	Likelihood	Impact	Mitigation
CORS restrictions block API Request Tester	High	Medium	Ship optional CORS proxy; document limitations clearly
Tesseract.js OCR slow on large images	Medium	Medium	Web Worker + progress bar; file size limit of 10 MB
Background removal WASM model too large (>40 MB)	Medium	Medium	Lazy load on first use; show download progress
Exchange rate API rate limits or downtime	Low	Low	1-hour localStorage cache; static fallback rates
Browser PDF print API inconsistencies across browsers	Medium	High	Use jsPDF as primary; browser print as fallback
LocalStorage quota exceeded for large datasets	Low	Medium	Warn user; offer JSON file export as alternative
Legal concerns about contract templates	Low	High	Prominent disclaimer: templates, not legal advice
SEO cannibalisation if tool descriptions are thin	Medium	Medium	Unique meta descriptions and structured tool content pages

11. Appendix

A. Tool Category Breakdown

Category	Count	Tools
Text	9	Text Pattern Extractor, Data Cleanup Suite, Bulk File Naming, Markdown Editor, Text Extractor, Text Intelligence Analyzer, CSV/Table Builder, Regex Builder (text output), Structured Comm. Builder
Planning	6	ATS Resume Builder, Proposal Generator, Timezone Overlap, FocusFlow, Capstone Idea Builder, UI System Blueprint, Contract Builder
Numbers	5	Pricing Calculator, Quotation Generator, Invoice Builder, Currency Converter, Food Costing Tool
Links	4	QR Code Builder, YouTube Video Inspector, cURL Converter, API Request Tester
Developer	6	Regex Builder, API Request Tester, cURL Converter, Nginx Config, .htaccess Generator, JWT Decoder, Diagram Builder, System Spec Builder
Media	7	Image Compressor, Social Post Mockup, Background Remover, Icon System Builder, Social Post Mockup, SVG System Builder, Sketch Tool, Color System Toolkit, ID Card Generator
PDF	4	Images to PDF, PDF to DOCX, Text Extractor, Bulk Certificate Generator

B. Glossary

- ATS:** Applicant Tracking System — software used by employers to filter resumes
- CSP:** Content Security Policy — HTTP header that controls resource loading
- CORS:** Cross-Origin Resource Sharing — browser security mechanism for cross-domain requests

JWT: JSON Web Token — compact, URL-safe authentication token format

LCP: Largest Contentful Paint — Core Web Vitals metric for perceived load speed

OCR: Optical Character Recognition — extracting text from images

WASM: WebAssembly — binary instruction format enabling near-native performance in browsers

WCAG: Web Content Accessibility Guidelines — international accessibility standard

Kit: A curated collection of related tools grouped for a specific workflow context

localStorage: Browser-native key-value storage persisted across sessions, scoped to origin